POWER BI

Power BI Interview Notes

The model, the DAX and the concepts that come up in every Power BI round.

- **Star schema, explained simply**
- **6 DAX measures you will reuse**
- **Top questions with answers**

Model it right

Most slow or wrong reports are a modelling problem, not a DAX problem.

- **Star schema**

  One fact table (Sales) in the middle. Dimension tables (Date, Product, Customer) around it.

- **One-to-many, single direction**

  Filters flow from dimension to fact. Avoid both-direction filters unless you must.

- **Always add a Date table**

  Mark it as a date table. Time intelligence functions need it.

```
Date = CALENDAR(DATE(2024,1,1),
DATE(2026,12,31))
```

6 measures you will reuse

Write these once and you can build most business dashboards.

```
Total = SUM(Sales[Amount])

North = CALCULATE([Total],
Region[Name]="North")

% of Total = DIVIDE([Total],
    CALCULATE([Total], ALL(Region)))

Sales LY = CALCULATE([Total],
    SAMEPERIODLASTYEAR('Date'[Date]))

YoY % = DIVIDE([Total] - [Sales LY],
[Sales LY])

YTD = TOTALYTD([Total], 'Date'[Date])
```

DIVIDE instead of "/" returns blank instead of an error when the bottom is 0.

Concepts they test

Know the one line answer, then give an example.

- **Measure vs calculated column**

  Column: computed row by row at refresh, stored, used for slicing. Measure: computed at query time, used for totals and KPIs.

- **Row context vs filter context**

  Row context is the current row. Filter context is what slicers, visuals and CALCULATE have filtered.

- **SUM vs SUMX**

  SUMX calculates row by row first, then adds. Use it for Qty × Price.

```
SUMX(Sales, Sales[Qty] * Sales[Price])
```

Most asked questions

Prepare a 2 line answer for each.

- **What does CALCULATE do?**
  Changes the filter context, then evaluates the expression.
- **Import vs DirectQuery**
  Import is fast, data is copied. DirectQuery is live, slower, for very large or real-time data.
- **Row Level Security**
  Roles with DAX filters so each manager sees only their region.
- **Slow report?**
  Fewer visuals per page, remove unused columns, measures over calculated columns, star schema.